

KRISTINE STANLEY

PYSPARK DATA ENGINEER

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PROFESSIONAL SUMMARY

Senior PySpark Data Engineer with 11+ years across cybersecurity, connected mobility, insurance, and learning platforms, delivering Spark 3.x, PySpark, Databricks, Delta Lake, Python 3.8-3.12, Airflow 2.x, modern cloud analytics platforms solutions with secure APIs, scalable architecture, cloud delivery, observability, and production modernization that improves reliability, performance, release quality, and business scalability for enterprise products.

SKILLS

- **Data Engineering:** Spark 3.x, PySpark, Databricks, Delta Lake, Python 3.8-3.12, Airflow 2.x, data quality, governance, orchestration, dashboards, cloud analytics, CI/CD, ETL/ELT, orchestration, warehouse modeling, streaming, data quality, governance, analytics enablement
- **Analytics & Modeling:** star schemas, dimensional modeling, semantic layers, dashboards, metrics definitions, lineage, SLAs, anomaly detection, data contracts
- **Architecture:** clean architecture, domain-driven design, microservices, event-driven architecture, modular monoliths, CQRS, transactional outbox, API versioning
- **Cloud & DevOps:** Azure, AWS, Docker, Kubernetes, Terraform, Helm, GitHub Actions, Azure DevOps, Jenkins, CI/CD, feature flags, blue-green deployments
- **Databases & Messaging:** PostgreSQL, SQL Server, Redis, MySQL, MongoDB, Kafka, RabbitMQ, Azure Service Bus, AWS SQS, indexing, migrations, query optimization, caching
- **Testing & Quality:** xUnit, NUnit, Playwright, Cypress, Jest, Testcontainers, Pact, integration testing, contract testing, load testing, accessibility testing, static analysis, secure code review
- **Observability & Leadership:** OpenTelemetry, Serilog, Datadog, Application Insights, Prometheus, Grafana, incident response, root-cause analysis, mentoring, architecture reviews, stakeholder communication

RELEVANT EXPERIENCE

Senior Software Engineer

April 2021 - May 2026

Veracode Systems Inc — Remote / Full time, Somerville, MA

- Led senior delivery for enterprise application-security products using **Spark 3.x, PySpark, Databricks, Delta Lake, Python 3.8-3.12, Airflow 2.x, Kafka, PostgreSQL, Azure, AWS, Docker, data quality checks, observability** across data ingestion pipelines, warehouse models, operational metrics, and analytics delivery workflows for security, product, platform, and support teams.
- Reorganized tightly coupled services into domain-focused modules using **clean architecture**, typed contracts, dependency injection, and reviewable service boundaries, reducing recurring production defects by **36%**.
- Resolved P95 latency caused by expensive joins, skewed partitions, duplicate loads, and unbounded reporting extracts through **query tuning**, caching, pagination, asynchronous processing, and payload shaping, improving critical workflow response time by **43%**.
- Modernized legacy components from older runtime patterns to **Spark / PySpark Data** practices, replacing obsolete packages, synchronous calls, unsafe shared state, and fragile deployment assumptions with maintainable release paths.
- Designed secure multi-tenant access patterns with **OAuth 2.0, OpenID Connect**, JWT validation, tenant-scoped data access, RBAC, audit logging, and policy-based authorization for sensitive customer data.
- Eliminated duplicate processing during retries by implementing **idempotency keys**, transactional state changes, outbox publishing, dead-letter replay procedures, and correlation identifiers across service boundaries.
- Built curated datasets, governed metrics, streaming alerts, self-service dashboards, and pipeline orchestration while coordinating backlog priorities, technical design, rollout planning, and dependency management with product managers, QA engineers, security reviewers, and customer-facing teams.
- Established automated quality gates with **data quality tests, schema contracts, pipeline integration tests, reconciliation checks, and lineage reviews**, catching contract mismatches, migration failures, data regressions, and release-blocking defects before production deployment.
- Introduced operational dashboards with **OpenTelemetry, Serilog, Datadog, Application Insights, Prometheus, Grafana**, accelerating diagnosis of memory leaks, queue backlogs, database contention, failed integrations, and downstream timeout incidents.

Software Engineer

January 2019 - April 2021

Cubic Telecom USA — Remote / Full time, San Diego, CA

- Developed connected-mobility platform services using **Spark 3.x, PySpark, Databricks, Delta Lake, Python 3.8-3.12, Airflow 2.x, PostgreSQL, RabbitMQ, Azure DevOps, scheduled ETL, reporting automation** for device registration, SIM provisioning, subscription changes, carrier synchronization, billing events, and customer operations.
- Improved provisioning throughput by **38%** through asynchronous command handling, bounded concurrency, batched persistence, rate limiting, and resilient retry policies that protected external telecom APIs.
- Resolved failed activations caused by inconsistent carrier identifiers, incompatible payload structures, and delayed callbacks by creating canonical models, provider adapters, and replayable reconciliation workflows.
- Designed versioned API contracts with **OpenAPI**, validation rules, standardized error responses, permission-aware endpoints, and documented integration examples for portals and operational tools.
- Implemented message-driven workflows with durable queues, dead-letter handling, consumer checkpoints, idempotent updates, and correlation identifiers to prevent duplicate activations and preserve traceability.
- Corrected stale device states caused by out-of-order events through timestamp validation, monotonic transition rules, scheduled reconciliation jobs, and immutable status histories for support teams.
- Optimized slow device, account, and usage queries using covering indexes, compiled projections, connection-pool tuning, and removal of N+1 access patterns as connected-device volumes increased.
- Standardized delivery with **Docker, Azure DevOps**, automated tests, security scans, migration validation, environment checks, approval gates, and rollback notes for distributed releases.

Software Developer

January 2017 - December 2018

Asurion — Remote / Full time, Nashville, TN

- Built insurance and service-contract workflows using **SQL Server, Python 3.5-3.7, SSIS-style jobs, Redis, reporting APIs, data validation** for policy eligibility, claim intake, payment callbacks, appointment scheduling, repair status, and customer communications.
- Increased successful online claim completion by **27%** after redesigning orchestration flows to preserve recoverable state, return actionable validation details, and reduce unnecessary calls to slow dependencies.
- Created a backend-for-frontend and integration layer that combined customer, policy, inventory, payment, and appointment data while shielding product experiences from incompatible downstream contracts.
- Corrected duplicate appointment and payment records caused by repeated submissions through idempotency tokens, uniqueness constraints, transactional validation, and request-state verification.
- Implemented caching for frequently requested eligibility, location, and reference data with explicit invalidation, short expiration periods, and monitoring to improve speed without serving stale decisions.
- Developed scheduled reconciliation services for incomplete claims and delayed provider callbacks, recovering transactions after timeout conditions while preserving audit records for service and finance teams.
- Expanded automated coverage with **NUnit, Moq**, Selenium, service virtualization, API integration tests, and failure-scenario suites for claims, payments, appointments, and eligibility decisions.

Junior Software Developer

June 2015 - December 2016

Rustici Software — Hybrid / Full time, Nashville, TN

- Maintained learning-platform applications using **SQL, MySQL, Python scripts, Excel exports, ETL validation, operational reporting automation** for course imports, learner-state persistence, launch configuration, completion tracking, reporting extracts, and xAPI-style event handling.
- Implemented administration workflows and integration endpoints for content ingestion, adding validation for malformed archives, incomplete manifests, missing metadata, and nonstandard customer payloads.
- Resolved excessive memory consumption during large imports by streaming compressed packages, limiting parallel extraction, disposing temporary resources, and recording actionable processing failures for support teams.
- Built defensive processing for duplicate and out-of-order learner events, preserving accurate completion, score, and resume state across platforms with inconsistent callback and retry behavior.
- Improved database reliability through targeted indexes, transactional updates, parameterized queries, and repeatable migration scripts for learner-state tables used during high-volume reporting periods.
- Collaborated with senior engineers and support specialists to reproduce unusual integration failures, document payload assumptions, and deliver maintainable fixes that shortened customer troubleshooting cycles.

SELECTED PROJECTS

- **Enterprise Data Pipeline Modernization:** Designed a governed pipeline platform with **Spark 3.x, PySpark, Databricks, Delta Lake, Python 3.8-3.12, Airflow 2.x, data quality, governance, orchestration, dashboards, cloud analytics, CI/CD**, schema contracts, incremental loads, lineage checks, and dashboard-ready marts that reduced repeated analyst rework and improved metric trust.

- **Analytics Reliability and Data Quality Platform:** Built a quality and observability framework with reconciliation tests, freshness SLAs, anomaly alerts, and runbooks for pipeline failures, late data, duplicate records, and warehouse cost spikes.

EDUCATION

**Tennessee State University — Bachelor's Degree in
Computer Science**

Sep 2011 - May 2015

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate
- Microsoft Certified: Azure Solutions Architect Expert
- AWS Certified Developer - Associate
- Professional Scrum Master I - PSM I